

Continuous assessment IV

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Question 1

We know $N_F = N_B$ With

$$N_i = z \frac{\partial}{\partial z} \ln \mathcal{Z} \quad (1)$$

$$\ln \mathcal{Z}_{\pm} = \frac{A}{\lambda_T^2} kT \mathcal{L}i_2(\pm z) \quad (2)$$

We derived the d-dim density of states here we are in two dimensional case which gives us our prefactor. We can let $N_F = N_B$ and use the common relation outlined below to get

$$-\frac{A}{\lambda_T^2} \mathcal{L}i_1(-z_F) = \frac{A}{\lambda_T^2} \mathcal{L}i_1(z_B) \quad (3)$$

Using the definition of the polylogarithm of order 1 $\mathcal{L}i_1(z) = -\ln(1-z)$. Thus we have

$$\ln(1+z_F) = -\ln(1-z_B) \quad (4)$$

$$z_B = \frac{z_F}{1+z_F} \quad (5)$$

We can use the common relation

$$z \frac{d}{dz} \mathcal{L}i_r(z) = \mathcal{L}i_{r-1}(z) \quad (6)$$

$$\frac{d}{dz} \mathcal{L}i_2(z) = \frac{\mathcal{L}i_1(z)}{z} \quad (7)$$

$$\frac{d}{dz} \mathcal{L}i_2(z) = -\frac{\ln(1-z)}{z} \quad (8)$$

We can integrate both sides up to the fermi-fugacity.

$$-\int_0^{-z_F} \frac{d}{dz} \mathcal{L}i_2(z) dz = \int_0^{-z_F} \frac{\ln(1-z)}{z} dz \quad (9)$$

$$-\mathcal{L}i_2(-z_F) = \int_0^{z_F} \frac{\ln(1+u)}{u} du \quad (10)$$

where $u = -z$. Now we can do partial fractions to make the integral nicer as we want to have a nice sub for the argument of the log.

$$\frac{1}{u} = \frac{1}{1+u} + \frac{1}{f(u)} \quad (11)$$

$$\implies f(u) = u(1+u) \quad (12)$$

$$-\mathcal{L}i_2(-z_F) = \int_0^{z_F} \frac{\ln(1+u)}{1+u} du + \int_0^{z_F} \frac{\ln(1+u)}{u(1+u)} du \quad (13)$$

For the first integral we can use $\tilde{u} = \ln(1+u)$. For the second one we can do some devious algebra in the hopes of writing in the form of a polylogarithm such that we can use the aforementioned relation.

$$-\mathcal{L}i_2(-z_F) = \frac{\ln^2(1+u)}{2} \Big|_0^{z_F} - \int_0^{z_F} \ln\left(\frac{1}{1+u}\right) \frac{1+u}{u} \frac{1}{(1+u)^2} du \quad (14)$$

Now we can use a sub so that we have the polylogarithm form. The sub $t = \frac{u}{1+u}$.

$$-\mathcal{L}i_2(-z_F) = \frac{1}{2} \ln^2(1-z_f) - \int_0^{\frac{z_F}{1+z_F}} \ln(1-t) \frac{dt}{t} \quad (15)$$

$$(16)$$

Using the relation we have

$$-\mathcal{L}i_2(-z_F) = \frac{1}{2} \ln^2(1 - z_F) - \int_0^{\frac{z_F}{1+z_F}} \frac{d}{dt} \mathcal{L}i_2(t) dt \quad (17)$$

$$= \frac{1}{2} \ln^2(1 - z_F) \mathcal{L}i_2(z_B) \quad (18)$$

Thus we get

$$\Rightarrow -\frac{AkT}{\lambda_T^2} \mathcal{L}i_2(-z_F) = \frac{AkT}{\lambda_T^2} \left[\frac{1}{2} \ln^2(1 - z_F) \mathcal{L}i_2(z_B) \right] \quad (19)$$

We can identify the fermi and Bose energy

$$E_F = E_B + \frac{kT}{2A} \lambda_T^2 \left[\frac{A}{\lambda_T^2} \ln(1 - z_F) \right]^2 \quad (20)$$

$$= E_B + \frac{kT}{2A} \frac{h^2}{2\pi m k T} [N^2] \quad (21)$$

$$= E_B + \frac{N^2 h^2}{4\pi m A} \quad (22)$$

Here we see that the energies only differ by a constant (term independent of T) Thus we have that

$$\left(\frac{\partial E_F}{\partial T} \right)_{V,N} = \left(\frac{\partial E_B}{\partial T} \right)_{V,N} \quad (23)$$

$$\Rightarrow C_{V,F} = C_{V,B} \quad (24)$$

We can look at the ground state energy. We have the following expressions

$$E_F(N, 0) = E_B(N, 0) + \frac{N^2 h^2}{4\pi m A} = \int_0^{\epsilon_f} W(1, \epsilon) \epsilon d\epsilon \quad (25)$$

$$0 + \frac{N^2 h^2}{4\pi m A} = \int_0^{\epsilon_f} \frac{2\pi m A}{h^2} \epsilon d\epsilon \quad (26)$$

$$\frac{N^2 h^2}{4\pi m A} = \frac{\pi m A \epsilon_F^2}{h^2} \quad (27)$$

$$\Rightarrow N = \frac{2\pi m A \epsilon_F}{h^2} \quad (28)$$

But we have

$$E_F(N, 0) = \frac{\pi m A \epsilon_F^2}{h^2} \quad (29)$$

So we can identify

$$E_F(N, 0) = \frac{1}{2} \epsilon_F \left(\frac{2\pi m A \epsilon_F}{h^2} \right) = \frac{1}{2} \epsilon_F N \quad (30)$$

Question 2

For a Fermi gas if we have $T = 0$. We have

$$N = \int_0^{\epsilon_f} d\epsilon W(1, \epsilon) d\epsilon \quad (31)$$

We want to extend this to the low non-zero temperature regime. We need to include the distribution

$$N = \int_0^{\epsilon_f} d\epsilon \frac{W(1, \epsilon)}{1 + e^{\beta(\epsilon - \mu)}} \quad (32)$$

Which leads to

$$\langle E \rangle = \int_0^{\epsilon_f} \frac{\epsilon W(1, \epsilon) d\epsilon}{1 + e^{\beta(\epsilon - \mu)}} \quad (33)$$

We have the approximation

$$\int_0^\infty dx \frac{f(x)}{1 + e^{x - \xi}} \approx \int_0^\xi dx f(x) + \frac{\pi^2}{6} \left(\frac{\partial f}{\partial x} \right)_{x=\xi} + \dots \quad (34)$$

we can let $f(x) = \frac{1}{\beta} W\left(\frac{x}{\beta}\right)$ with $x = \beta\epsilon$.

Thus we get

$$N = \int_0^\infty \frac{W\left(\frac{x}{\beta}\right)}{1 + e^{x - \beta\mu}} d\left(\frac{x}{\beta}\right) \approx \int_0^{\beta\mu} dx W\left(\frac{x}{\beta}\right) + \frac{1}{\beta} \frac{\pi^2}{6} \left(\frac{\partial W\left(\frac{x}{\beta}\right)}{\partial x} \right)_{x=\beta\mu} \quad (35)$$

$$\approx \int_0^\mu W(1, \epsilon) d\epsilon + \frac{1}{\beta^2} \frac{\pi^2}{6} \left(\frac{\partial W(1, \epsilon)}{\partial \epsilon} \right)_{\epsilon=\epsilon_f} \quad (36)$$

$$N = \int_0^\mu W(1, \epsilon) d\epsilon + \frac{\pi^2}{6} (kT)^2 \left(\frac{dW(1, \epsilon)}{d\epsilon} \right)_{\epsilon=\mu} \quad (37)$$

As we are in the small temperature limit $\implies Z \gg 1$ ($z = e^{\beta\mu}$) and $\mu \approx \epsilon_f$.

Thus we get

$$N = \int_0^{\epsilon_f} W(1, \epsilon) d\epsilon - \Delta W(1, \epsilon) + (\dots)_{\epsilon=\mu} \quad (38)$$

where we define $\Delta = \epsilon_f - \mu$. Here we can identify that the first term to be integrated is actually just the total number of particles N so we get

$$N = N - (\epsilon_f - \mu) W + (\dots)_{\mu=\epsilon_f} \quad (39)$$

$$0 = -(\epsilon_f - \mu) W + \dots \quad (40)$$

$$\implies \mu = \epsilon_f - \frac{\pi^2}{6} \frac{(kT)^2}{W} \left(\frac{\partial W}{\partial \epsilon} \right)_{\epsilon=\epsilon_f} \quad (41)$$

$$\implies \mu = \epsilon_f \left[1 - \frac{\pi^2}{6} \left(\frac{\partial \ln(W(1, \epsilon))}{\partial \ln \epsilon} \right)_{\epsilon=\epsilon_f} \left(\frac{kT}{\epsilon_f} \right)^2 \right] \quad (42)$$

For the energy we have

$$\langle E \rangle = \int_0^\infty d\epsilon \frac{\epsilon W(1, \epsilon)}{1 + e^{\beta(\epsilon - \mu)}} \quad (43)$$

We can sub in $x = \beta\epsilon$ and $\xi = \beta\mu$

$$\frac{1}{\beta} \int_0^\infty \frac{\frac{x}{\beta} W\left(\frac{x}{\beta}\right) dx}{1 + e^{x - \beta\mu}} \approx \frac{1}{\beta} \int_0^{\beta\mu} \frac{x}{\beta} W\left(\frac{x}{\beta}\right) dx + \frac{1}{\beta} \frac{\pi^2}{6} \frac{\partial}{\partial x} \left[\frac{x}{\beta} W\left(\frac{x}{\beta}\right) \right] \quad (44)$$

$$\approx \frac{1}{\beta} \int_0^\mu \epsilon W(\epsilon) d\epsilon + \frac{1}{\beta^2} \frac{\pi^2}{6} \frac{\partial}{\partial \epsilon} [\epsilon W(\epsilon)] \quad (45)$$

$$(46)$$

We can taylor expand

$$\int_0^{\epsilon_f} \epsilon W(\epsilon) d\epsilon + \epsilon_f W(\epsilon_f) (\mu - \epsilon_f) + \dots \quad (47)$$

$$\approx E_0 - \epsilon_f^2 W(\epsilon_f) + \epsilon_f W(\epsilon_f) \left[\epsilon_f \left[1 - \frac{\pi^2}{6} \left(\frac{\partial \ln(W(1, \epsilon))}{\partial \ln \epsilon} \right)_{\epsilon=\epsilon_f} \left(\frac{kT}{\epsilon_f} \right)^2 \right] \right] + \frac{\pi^2}{6\beta^2} W(\epsilon_f) + \frac{\pi^2 \epsilon}{6\beta^2} \left(\frac{\partial W}{\partial \epsilon} \right)_{\epsilon=\mu} \quad (48)$$

$$\langle E \rangle \approx E_0 - \epsilon_f^2 W(\epsilon_f) + \epsilon_f^2 W(\epsilon_f) - \frac{\epsilon_f^2 W(\epsilon_f) \pi^2}{6} \left(\frac{\epsilon_f}{W} \left(\frac{\partial W}{\partial \epsilon} \right)_{\epsilon=\epsilon_f} \right) \left(\frac{kT}{\epsilon_f} \right)^2 + \frac{\pi^2}{6\beta^2} W(\epsilon_f) + \frac{\pi^2 \epsilon}{6\beta^2} \left(\frac{\partial W}{\partial \epsilon} \right)_{\epsilon=\mu} \quad (49)$$

$$\langle E \rangle \approx E_0 + \frac{\pi^2 W(\epsilon_f)}{6\beta^2} \quad (50)$$

We obtain

$$E \approx E_0 - \frac{\pi^2}{6} k^2 T^2 W(1, \epsilon_f) \quad (51)$$

$$C_V = \frac{\partial E}{\partial T} \quad (52)$$

$$C_V = \frac{\pi^2}{3} k^2 T W(1, \epsilon_f) \quad (53)$$

Given that we have $\epsilon_i \propto p^s$

$$W(\epsilon) = \delta(\epsilon) + (deg) \int_{-\infty}^{\infty} \delta(\epsilon - \epsilon_i) d^d i \quad (54)$$

$$= \delta(\epsilon) + (deg) \int_{-\infty}^{\infty} \delta(\epsilon - cp^s) d^d p \quad (55)$$

$$= \delta(\epsilon) + (deg) \frac{L^d}{h^d} S_{d-1}(\deg) \int_0^{\infty} |p|^{d-1} \delta(\epsilon - c|p|^s) d^d |p| \quad (56)$$

$$= \delta(\epsilon) + (deg) \frac{L^d}{h^d} S_{d-1}(\deg) \frac{1}{S} \frac{1}{c^{\frac{1}{s}}} \int_0^{\infty} (|p|)^{d-1} \delta(\epsilon - c|p|^s) d|p| \quad (57)$$

$$= \delta(\epsilon) + (deg) \frac{L^d}{h^d} S_{d-1}(\deg) \frac{1}{S} \frac{1}{c^{\frac{1}{s}}} \int_0^{\infty} \left(\frac{t}{c} \right)^{\frac{d}{s}-\frac{1}{s}} \delta(\epsilon - t) t^{\frac{1}{s}-1} dt \quad (58)$$

$$W(\epsilon) = \delta(\epsilon) + (deg) \frac{L^d}{h^d} S_{d-1}(\deg) \left(\frac{\epsilon}{c} \right)^{\frac{d}{s}-\frac{1}{s}} \frac{1}{S} \frac{1}{c^{\frac{1}{s}}} \epsilon^{\frac{1}{s}-1} \quad (59)$$

$$W(\epsilon) = \delta(\epsilon) + \tilde{c} \epsilon^{\frac{d}{s}-1} \quad (60)$$

where $\tilde{c}$ is a constant. For low temps we obtain

$$\mu = \epsilon_f - \frac{\pi^2}{6} \frac{(kT)^2}{W} \left(\frac{\partial W}{\partial \epsilon} \right)_{\epsilon=\epsilon_f} \quad (61)$$

$$\Rightarrow \mu = \epsilon_f - \frac{\pi^2}{6} \frac{(kT)^2}{W} \tilde{c} \left(\frac{d}{s} - 1 \right) \epsilon_f^{\frac{d}{s}-1} \quad (62)$$

$$C_V = \frac{\pi^2}{3} k^2 T \left(\tilde{c} \epsilon_f^{\frac{d}{s}-1} \right) \quad (63)$$

$$E \approx E_0 - \frac{\pi^2}{6} k^2 T^2 \left(\tilde{c} \epsilon_f^{\frac{d}{s}-1} \right) \quad (64)$$

$$S = \int_0^T C_V \frac{dT}{T} \quad (65)$$

$$S = \int_0^T \frac{\pi^2}{3} k^2 T W(1, \epsilon_f) \frac{dT}{T} \quad (66)$$

$$\Rightarrow S \approx C_V \approx \frac{\pi^2}{3} T W(1, \epsilon_f) \quad (67)$$

We have

Question 3

We are given

$$\sum_i n_i \epsilon_i = E \quad (68)$$

$$\sum_i n_i = N \quad (69)$$

We know that

$$\Omega = \sum_{\{n_i\}}^\infty W_{\{n_i\}} \quad (70)$$

$$W_{\{n_i\}} = \prod w(i) \quad (71)$$

For the Bose-Einstein case we know

$$w(i) = \frac{(n_i + g_i - 1)!}{n_i! (g_i - 1)!} \quad (72)$$

We have

$$S(N, V, E) = k \ln \Omega = k \ln \left(\prod_{\{n_i\}} W(n_i) \right) = \sum_{\{n_i\}} \ln w(i) \quad (73)$$

We can take the approximation that $\sum_i n_i = n_i^*$.

Thus we have

$$S(N, V, E) = \sum_{\{n_i\}} k \ln \left(\frac{(n_i^* + g_i - 1)!}{n_i^*! (g_i - 1)!} \right) = \sum_{\{n_i\}} k [\ln(n_i^* + g_i - 1)! - \ln(n_i^*!) - \ln((g_i - 1)!)] \quad (74)$$

$$\sum_{\{n_i\}} k [(n_i^* + g_i - 1) \ln(n_i^* + g_i - 1) - (n_i^* + g_i - 1) - n_i^* \ln(n_i^*) + n_i^* - (g_i - 1) \ln(g_i - 1) + (g_i - 1)] \quad (75)$$

where we have used Stirlings approximation

$$\ln N! \approx N \ln N - N \quad (76)$$

We have $g_i \gg 1$

$$\sum_{\{n_i\}} k [(n_i^* + g) \ln(n_i^* + g) - (n_i^* + g) - n_i^* \ln(n_i^*) + n_i^* - g \ln(g) + g] \quad (77)$$

$$\sum_{\{n_i\}} k [(n_i^* + g) \ln(n_i^* + g) - n_i^* \ln(n_i^*) - g \ln(g)] \quad (78)$$

Now we can use the principle of min entropy using the fact that we have E and N fixed so we can use Lagrange multipliers. We define $\mathcal{F} = \frac{S}{k}$ so $\mathcal{F} = \ln \Omega$ as we know that entropy is given by $S = k \ln \Omega$. We define our function with lagrange multipliers α and β .

$$\mathcal{F} = \sum_{\{n_i\}} [\ln \Omega_i - \alpha n_i - n_i \epsilon_i] \quad (79)$$

Since we want to find the minima we have

$$\delta \mathcal{F} = \delta \left(\sum_{\{n_i\}} [\ln \Omega_i - \alpha n_i - n_i \epsilon_i] \right) = 0 \quad (80)$$

$$\frac{d}{dn_i} [[(n_i * + g) \ln (n_i * + g) - n_i * \ln (n_i *) - g \ln (g)] - \alpha n_i - \beta n_i \epsilon_i] = 0 \quad (81)$$

$$\implies [\ln (n_i * + g) + 1 - \ln (n_i *) - 1] - \alpha - \beta \epsilon_i = 0 \quad (82)$$

$$\ln \left(\frac{g_i}{n_i *} + 1 \right) - \alpha - \beta \epsilon_i = 0 \quad (83)$$

$$\ln \left(\frac{g_i}{n_i *} + 1 \right) = \alpha + \beta \epsilon_i \quad (84)$$

$$\frac{g_i}{n_i *} = e^{(\alpha + \beta \epsilon_i)} - 1 \quad (85)$$

$$n_i * = \frac{g_i}{e^{(\alpha + \beta \epsilon_i)} - 1} \quad (86)$$

We can compare to the result of the average occupancy number for bosons which I will derive now

$$\mathcal{Z} = \sum_{\{n_i\}} e^{-\beta \sum_i n_i (\epsilon_i - \mu)} \quad (87)$$

$$\mathcal{Z} = \prod_i \mathcal{Z}_i \quad (88)$$

$$\mathcal{Z}_i = \sum_{n_i} e^{-\beta n_i (\epsilon_i - \mu)} \quad (89)$$

$$(90)$$

using the geometric series

$$\mathcal{Z}_i = \frac{1}{1 - e^{-\beta(\epsilon_i - \mu)}} \quad (91)$$

We can use the average of a quantity to find

$$\langle n_i \rangle = \frac{1}{\mathcal{Z}_i} \sum_{n_i=0}^{\infty} n_i e^{-\beta n_i (\epsilon_i - \mu)} \quad (92)$$

given the standard geometric series ($x = e^{-\beta(\epsilon_i - \mu)}$) we can take a derivative and then multiply by x to get

$$\sum_{n=0}^{\infty} n x^n = \frac{x}{(1-x)^2} \quad (93)$$

$$\implies \langle n_i \rangle = \frac{1}{\mathcal{Z}_i} \frac{x}{(1-x)^2} = (1-x) \frac{x}{(1-x)^2} = \frac{x}{1-x} \quad (94)$$

$$\langle n_i \rangle = \frac{g_i}{e^{\beta(\epsilon_i - \mu)} - 1} \quad (95)$$

We can identify from the BE case

$$\beta = \beta = \frac{1}{kT} \quad (96)$$

$$\alpha = -\beta\mu = -\frac{\mu}{kT} \quad (97)$$

For the Fermi case we have

$$\omega = \frac{g_i!}{n_i!(g_i - n_i)!} \quad (98)$$

$$\ln \omega = \ln(g_i!) - \ln(n_i!) - \ln(g_i - n_i)! \quad (99)$$

Using Stirling's approximation

$$\ln \omega(i) \approx g_i \ln g_i - g_i - n_i \ln n_i + n_i - (g_i - n_i) \ln(g_i - n_i) + g_i - n_i \quad (100)$$

$$\ln(\omega(i)) \approx \ln g_i - n_i \ln n_i - (g_i - n_i) \ln(g_i - n_i) \quad (101)$$

$$\frac{d}{dn_i} \mathcal{F} \approx -\ln n_i - 1 + \ln(g_i - n_i) + 1 - \alpha - \beta\epsilon_i = 0 \quad (102)$$

$$\ln\left(\frac{g_i}{n_i} - 1\right) - \alpha - \beta\epsilon_i = 0 \quad (103)$$

$$\frac{g_i}{n_i} = 1 + e^{\alpha + \beta\epsilon_i} \quad (104)$$

$$n_i = \frac{g_i}{1 + e^{\alpha + \beta\epsilon_i}} \quad (105)$$

We can compare this to the average occupancy number for Fermions which I will derive now. Since the pauli exclusion principle only allows $n_i \in \{0, 1\}$.

$$\mathcal{Z}_i = 1 + e^{-\beta(\epsilon_i - \mu)} \quad (106)$$

$$\langle n_i \rangle = \frac{1}{\mathcal{Z}_i} \sum_{n_i=0}^1 n_i e^{-\beta n_i(\epsilon_i - \mu)} = \frac{0 \cdot 1 + 1 \cdot e^{-\beta(\epsilon_i - \mu)}}{1 + e^{-\beta(\epsilon_i - \mu)}} = \frac{1}{e^{\beta(\epsilon_i - \mu)} + 1} \quad (107)$$

We see we get the same results

$$\alpha = -\beta\mu = -\frac{\mu}{kT} \quad (108)$$

$$\beta = \frac{1}{kT} \quad (109)$$

Question 4

0.0.1 (a)

We know the following holds

$$\langle N \rangle = \int_0^\infty \frac{W(1, \epsilon)}{\frac{e^{\beta\epsilon}}{z} + 1} d\epsilon \quad (110)$$

$$= \frac{1}{2(\hbar\omega_0)^3} \int_0^\infty \frac{\epsilon^2}{\frac{e^{\beta\epsilon}}{z} + 1} d\epsilon \quad (111)$$

We can use a u sub $u = \beta\epsilon$

$$\langle N \rangle = \frac{1}{2(\beta\hbar\omega_0)^3} \frac{1}{\beta^3} \int_0^\infty \frac{u^2}{\frac{e^u}{z} + 1} \quad (112)$$

$$= -\frac{\Gamma(3)}{2(\beta\hbar\omega_0)^3} \frac{1}{\Gamma(3)} \int_0^\infty \frac{u^{3-1}}{\frac{e^u}{z} - 1} du \quad (113)$$

Thus we get

$$\langle N \rangle = -\frac{\Gamma(3)}{(\beta \hbar \omega_0)^3} \mathcal{L}i_3(-z) \quad (114)$$

Similarly we have a formula for the average energy

$$\langle E \rangle = \int_0^\infty \frac{W(1, \epsilon) d\epsilon}{\frac{e^{\beta \epsilon}}{z} + 1} \quad (115)$$

$$= \frac{1}{2(\hbar \omega_0)^3} \int_0^\infty \frac{\epsilon^3}{\frac{e^{\beta \epsilon}}{z} + 1} d\epsilon \quad (116)$$

$$= -\frac{\Gamma(4)}{2\beta^4 (\hbar \omega_0)^3 \Gamma(4)} \int_0^\infty \frac{u^{4-1}}{\frac{e^u}{-z} - 1} du \quad (117)$$

We get

$$\langle E \rangle = -\frac{\Gamma(4)}{2\beta^4 (\hbar \omega_0)^3} \mathcal{L}i_4(-z) \quad (118)$$

We have the following expression for the number of particles

$$N = \int_0^{\epsilon_f} W(1, \epsilon) d\epsilon \quad (119)$$

$$= \frac{1}{2(\hbar \omega_0)^3} \int_0^{\epsilon_f} \epsilon^2 d\epsilon \quad (120)$$

$$= \frac{\epsilon_f^3}{6(\hbar \omega_0)^3} \quad (121)$$

$$\implies \epsilon_f = \hbar \omega_0 (6N)^{\frac{1}{3}} \quad (122)$$

We have the following

$$N = \langle N \rangle \quad (123)$$

$$\implies \frac{\epsilon_f^3}{6(\hbar \omega_0)^3} = \int_0^\infty \frac{W(1, \epsilon)}{\frac{e^{\beta \epsilon}}{z} + 1} d\epsilon \quad (124)$$

$$(125)$$

using $T_f = \frac{\epsilon}{k}$

$$\frac{1}{2(\hbar \omega_0)^3} \int_0^\infty \frac{\epsilon^2}{e^{\beta \epsilon - \beta \mu} + 1} d\epsilon = \frac{(kT_f)^3}{6(\hbar \omega_0)^3} \quad (126)$$

Let $x = \beta \mu$. Rearranging we get

$$\frac{3}{(kT_f)^3} \frac{1}{\beta^3} \int_0^\infty \frac{x^2}{e^{x-\beta \mu} + 1} dx = 1 \quad (127)$$

$$\implies 3 \left(\frac{T}{T_f} \right)^3 \int_0^\infty \frac{x^2}{e^{x-\beta \mu} + 1} dx = 1 \quad (128)$$

As required.